Nano Batch Studio

The simple guide to making product images in batches

What this app does

Nano Batch Studio places your product photos into finished, styled scenes — many at a time, from a single prompt. It can also build images from a list of product names, or combine several products into one room.

In one sentence

Put photos in a folder, write one description of the scene you want, press Generate, and collect the finished images from another folder.

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Setting up for the first time

You only do this once. It takes about ten minutes.

1 Install Python

Go to **python.org/downloads** and install Python 3.10 or newer.

On Windows this matters: during installation, tick the box that says **Add Python to PATH**. If you miss it, the app won't start.

2 Put the app folder somewhere easy

Your Desktop or Documents is fine. The folder is called **nano-batch-studio** and contains a file named **nano_batch.py**.

3 Open a command window inside that folder

Open the folder, click once on the address bar at the top (the strip showing the folder path), type **cmd**, and press Enter. A black window opens, already pointed at your folder.

On a Mac, open Terminal, type **cd** followed by a space, then drag the folder onto the window and press Enter.

4 Install the extra parts

In that black window, type the line below and press Enter. Wait for it to finish — you'll see 'Successfully installed'.

```
pip install -r requirements.txt
```

5 Get an API key

The app needs a key to talk to the image service. Get one from **aistudio.google.com/apikey** (for the Nano Banana models) or **platform.openai.com/api-keys** (for the GPT Image models).

These services are paid, so you'll need billing switched on. Keep your key private — anyone with it can spend your credit.

6 Start the app

Double-click **run_batch.bat**. (On Mac or Linux, type **python nano_batch.py** in the terminal.) Paste your key into the **API key** box — the app remembers it, so you won't need to do this again.

A tour of the window

Three pages down the left, your settings in the middle, and the Generate button always at the bottom.

The three pages

Page	What it's for
Workspace	Where you do the work: pick your source, write the prompt, choose the model and output settings.
Product library	Optional. Lets the app dress your scenes using only your own furniture and decor, instead of inventing its own.
Activity	The running log — what's being made, what was saved, and any problems. Check here if something looks wrong.

The Workspace, step by step

Section	What you set
01 Choose your source	Which of the three ways you're working, and the folder or product list to work from.
02 Describe the scene	Your prompt — the description of the scene you want.
Generation model	Which image model to use, and the key for it.
Output settings	Aspect ratio, resolution, quality and file format.
Save your images	Where the finished images go.

Always at the bottom

- **Generate images** — starts the batch.
- **Stop** — halts after the current image; already-finished images are kept.
- The **progress bar** and a short status line show how far along you are.

Good to know Every setting you change is remembered automatically, so the app opens next time exactly as you left it. **Reset settings** in the bottom-left puts everything back to the defaults.

The three ways to work

Pick one at the top of the Workspace page. The fields you don't need are hidden automatically.

Photo scenes

One photo in, one image out. The app takes every photo in your input folder and builds a separate scene around each one. Your product stays the hero of its own image, and the finished file keeps the same filename as the original.

Use this when you have real product photos and want each one styled into a room. This is the most common choice.

Product list

No photos needed. The app reads product names from a CSV file and invents an image for each one from your description alone.

Your CSV needs a column headed **product**, and can have an optional **extra** column for per-product details. Write **{product}** in your prompt and each name is slotted in.

Combine

Many products, one room. The app takes several photos from your input folder and blends them into a single scene, then repeats to give you different versions.

Scene variations sets how many rooms you get. **Photos per scene** sets how many products go into each one — a number, or the word **all**.

If you want to…	Choose
Style each product in its own room	Photo scenes
Create images without having any photos	Product list
Furnish one room with several of your products	Combine

Making your first batch

Start small — two or three photos — so you can judge the result before running everything.

1 **Add your photos**
Copy two or three product photos into the **inputs** folder inside the app folder. JPG, PNG and WebP all work.

2 **Choose Photo scenes**
On the Workspace page, under **01 Choose your source**, click **Photo scenes**. The line underneath confirms what it will do.

3 **Write your prompt**
Under **02 Describe the scene**, describe the room you want. Be specific about style, light, materials and mood. For example:

> *Create a minimalist Scandinavian interior scene using this furniture piece as the hero product. Warm natural light, light wood floors, soft neutral textiles. A beautiful editorial photograph from a high-end design magazine. Keep the product's proportions unchanged.*

4 **Pick your settings**
Choose a model, then an aspect ratio (**3:2** suits editorial interiors), a resolution (**2K** is a good balance) and a file format.

5 **Press Generate images**
Watch the progress bar. Switch to the **Activity** page to see each file as it's saved.

6 **Collect your images**
Press **Open output folder**. Your finished images are there, named after the originals. Nothing is ever overwritten — a second run adds **-v2** to the name.

Tip If the result isn't right, change one thing at a time — first the prompt, then the model. Changing several at once makes it hard to tell what helped.

Using only your own furniture

By default the model invents furniture to fill the room. This stops it, and makes it use your catalogue instead.

Why you'd use it

Left alone, the model dreams up sofas, lamps and rugs that don't exist in your range. If you switch this on and give it a folder of your own product photos, it will only use those to dress the scene — choosing whichever suit and leaving the rest out.

How to switch it on

1. **Go to the Product library page**
   Click **Product library** in the left sidebar, then tick **Use only my own products to furnish the scene**.

   This works with **Photo scenes** mode only.

2. **Point it at your products**
   Choose **A folder on my computer** and browse to the folder holding your product photos. The app confirms how many it found.

3. **Set how many to offer**
   Supporting items per scene decides how many of your products the model gets to choose from for each room. Six is a good start. Fewer items usually means each one is reproduced more faithfully. The most it can use is 13.

Fetching from a website instead

Choose **A website**, paste the address of a product listing page, and press **Fetch images**. The app downloads the product photos into a local folder, which it then uses as the library.

- Open that folder afterwards and delete anything that isn't a real product — one bad reference can spoil a whole scene.
- Some shops load their images with JavaScript, which the app can't read. If nothing comes back, save the images to a folder by hand instead.
- Only use images you have the rights to.

Choosing a model, size and quality

Ten models from two companies. They all do the same job with different trade-offs.

Which model?

Model	Best for
Nano Banana Pro	Google's highest quality. A safe default for final work.
Nano Banana 2	Faster and cheaper than Pro. Good for large batches.
Nano Banana 2 Lite	Fastest and cheapest. Good for rough drafts.
GPT Image 2.5 Sunburst	Best at keeping your product exactly as it is. Slower. Strong choice for hero product work.
GPT Image 2.5 Flare	The same family but quicker. Good everyday choice.
Older GPT Image models	Kept for compatibility. They cannot do 2K or 4K.

Aspect ratio and resolution

- **Aspect ratio** is the shape. **3:2** suits editorial interiors, **4:5** suits Instagram, **16:9** suits banners.
- **Resolution** is the size: **1K** for quick drafts, **2K** for most work, **4K** for print.
- **Quality** only applies to the GPT Image models, and higher settings cost more.

The line under these settings always tells you the exact pixel size you'll get, so there's no guesswork.

Why 4K isn't always 3840 pixels wide OpenAI limits how many pixels an image can have in total. At 3:2 the largest allowed is 3520 x 2352. If you want a true 3840 x 2160, choose the 16:9 ratio at 4K.

Keeping costs sensible

- Test your prompt on two or three products before running the whole set.
- Draft at 1K, then re-run the ones you like at 2K or 4K.
- **Images per item** multiplies the cost — three images per product means three times the spend.

When something goes wrong

The Activity page explains most problems in plain language. Here are the common ones.

What you see	What to do
Double-clicking **run_batch.bat** does nothing	Python probably isn't on your PATH. Reinstall Python and tick **Add Python to PATH**. To see the real error, open a command window in the folder and type **python nano_batch.py**.
'openai' isn't installed yet (or 'google-genai')	Open a command window in the app folder and run **pip install -r requirements.txt**. Then close the app and open it again — it won't notice new parts until it restarts.
Your organization must be verified	OpenAI requires ID verification before it allows the GPT Image models. Go to platform.openai.com, then Settings, Organization, General. Access takes about 15 minutes to activate. Make sure your key belongs to the organisation you verified.
That API key was rejected	The key was mistyped, or belongs to the other provider. Each provider has its own key box — the label above it tells you which one is expected.
No images found in the input folder	The folder path is wrong, or the files aren't JPG, PNG or WebP. The line under the folder box tells you how many images were found.
Nothing downloads from a website	That shop probably loads images with JavaScript. Lower the minimum image size, or save the images to a folder by hand.
The model changed my product	Switch on **Use only my own products**, offer fewer supporting items, and add a line to your prompt such as 'keep the product's shape, colour and proportions unchanged'. GPT Image 2.5 Sunburst is the strongest at this.
It's slower than expected	Higher resolutions and quality settings take longer. **Stop** halts the run immediately and keeps whatever has already been saved.

Two things to remember

Your images are never overwritten — a repeat run adds **-v2** to the filename. And every setting is saved automatically, so you can close the app at any time without losing your setup.

Images made with Google's models carry an invisible SynthID watermark identifying them as AI-generated. This is built into the model, not added by this app.